# ALFA RIZI

SOFTWARE ENGINEER | GO, JAVA & FULL STACK

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## PROFESSIONAL SUMMARY

Software Engineer (Informatics Management D3 student, GPA 3.78/4.00; coursework completed, preparing for final thesis defense) with proven experience engineering distributed microservices, enterprise modular monoliths, AI/RAG pipelines, and interactive web platforms. Hands-on experience in Go (net/http, pgx), Dart/Flutter, SQLite, Java (Quarkus 3.15, Spring Boot 3.2, Spring Cloud Gateway), TypeScript/JavaScript (Next.js 14/16, React 19), Python (FastAPI, ChromaDB), and PostgreSQL (PL/pgSQL triggers, views, Flyway), with Docker containerization and automated GitHub Actions CI/CD.

## TECHNICAL SKILLS

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| **Languages:** | Java 17, TypeScript, JavaScript (ES6+), Python 3.12, SQL, C |
| **Backend & Systems:** | Quarkus 3.15 LTS, Spring Boot 3.2, Spring Cloud Gateway, FastAPI, Express.js, Double-Entry Escrow |
| **Frontend Development:** | Next.js 14/16 (App Router), React 18/19, Tailwind CSS v4, Web Speech API, Web Audio API, Canvas API |
| **Databases & Cache:** | PostgreSQL (Triggers, Views, Functions), Redis 7, Flyway Migrations, Drizzle ORM, ChromaDB |
| **DevOps & Tooling:** | Docker, Docker Compose, GitHub Actions (CI/CD), ArchUnit, Testcontainers, Git, Linux CLI, OpenAPI, IntelliJ IDEA |
| **Go & Mobile Development:** | Go (Golang), Dart, Flutter, net/http, pgx, PostgreSQL, SQLite (sqflite) |

## FEATURED ENGINEERING PROJECTS

**Catat Uang - Personal Finance Application** | ***Go, Dart, Flutter, PostgreSQL, SQLite*** 2026

- Developed an offline-capable Flutter app with transactions, bills, SQLite storage, and local reminders.
- Built a Go REST API with PostgreSQL, registration/login, revocable sessions, transaction/bill CRUD, and idempotent payments.
- Implemented opt-in transaction/bill sync with SQLite outboxes, idempotent retries, conflict review, encrypted backups and account recovery; backend/Android CI passed.

**ReLoop — Enterprise Circular-Commerce Platform & Escrow Architecture** | ***Java 17, Quarkus 3.15 LTS, Next.js 14, PostgreSQL 16, Redis 7*** 2026
Demo: https://reloop.biz.id | Code: github.com/a1fariz/ReLoop

- Architected a Quarkus 3.15 modular monolith across 25 bounded contexts with ArchUnit boundary verification and transactional outbox worker.
- Engineered a double-entry escrow ledger (DR=CR) and 15-minute anti-hoarding checkout leases with pessimistic locks (SELECT ... FOR UPDATE).
- Integrated Next.js 14 frontend with TanStack Query and Redis catalog caching, achieving low latency and zero inventory race conditions.

**ApexGrid — F1 Race Ticket Reservation & Microservices Platform** | ***Java 17, Spring Boot 3.2, Spring Cloud Gateway, PostgreSQL, React 18, Docker*** 2026
Code: github.com/a1fariz/ApexGrid

- Architected a 3-service distributed system (Gateway, User, Race) with Spring Cloud Gateway and stateless JWT RBAC security.
- Synchronized ticket inventory via PostgreSQL PL/pgSQL triggers with an automated 30-minute reservation expiry worker.
- Built an interactive React 18 admin analytics panel with OpenAPI/Swagger docs; containerized with Docker Compose.

**StudyMate AI — Asynchronous RAG Document Knowledge Assistant** | ***Python 3.12, FastAPI, ChromaDB, Google Gemini 2.0 Flash, Docker*** 2026
Code: github.com/a1fariz/StudyMateAI

- Built an asynchronous RAG pipeline parsing textbook PDFs via PyMuPDF with ChromaDB vector search and exact page citations.
- Orchestrated Gemini 2.0 Flash for automated summaries, quizzes, and 3D flashcards; enforced a 60 req/min token-bucket rate limiter.

**Vanguard IELTS — Academic Preparation & SRS Platform** | ***React 19, TypeScript, Vite, Tailwind CSS v4, Web Speech API*** 2026
Demo: https://vanguard-ielts.vercel.app | Code: github.com/a1fariz/Vanguard-IELTS

- Developed a 4-skill IELTS preparation suite (Listening, Reading, Writing, Speaking) covering CEFR A1 to C1 proficiency levels.
- Implemented a SuperMemo-inspired Spaced Repetition (SRS) vocabulary engine and Shadowing Studio with native Web Speech synthesis.

## PROFESSIONAL EXPERIENCE

**Lead Intern — IT & Business Operations** 2025 – Sep 2026
*PT Resika Cemerlang (Pasim Group)* *Bandung, Indonesia*

- Directed a 5-person intern team managing day-to-day IT support operations, hardware diagnostics, and workstation SLA ticket compliance.
- Enforced OS security patching, software configurations, and antivirus definition updates across corporate laptops.
- Validated technical specifications across thousands of medical device asset entries, ensuring database integrity and audit readiness.

## EDUCATION & TRAINING

**Universitas Nasional PASIM** Coursework completed; preparing for final thesis defense
*Associate Degree in Informatics Management (D3) — GPA: 3.78 / 4.00* *Bandung, Indonesia*

- PUB Intensive Technical Scholarship (2024 – Present): Logic & C, Data Structures, Relational Databases, Advanced Java Development, and React Development.
- Intensive React Front-End Development Training (Sep 2026); Fundamental Back-End Programming with Java and Git & GitHub Training (Jan 2026).
- Web Programming Training (Aug 2025); Database and Data Structure Training (May 2025).

## CERTIFICATIONS & CREDENTIALS

**Python for Everybody Specialization** — University of Michigan (2026)
**Certified Entry-Level Python Programmer (PCEP)** — Python Institute (2025)
**Programming for Data Science with Python** — HarvardX (2025)
**Introduction to Cloud Computing** — IBM (2026)